

MAHLANGU SAM BANELE

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PROFESSIONAL SUMMARY

Geomatics postgraduate with training in GIS, surveying, spatial data analysis and geospatial technologies, with growing interest in spatial intelligence, infrastructure systems and GeoAI driven workflows. Strong analytical foundation in mapping, spatial reasoning and data focused problem solving.

Eager to collaborate with diverse teams, embrace new challenges, and apply analytical thinking to deliver effective solutions. My goal is to grow as a versatile professional who can adapt to evolving technologies and contribute value across various sectors.

EDUCATION

Postgraduate Diploma in Geomatics

Tshwane University of Technology | 2025- Completed (2025)

Bachelor of Geomatics

Tshwane University of Technology | 2022- Completed (2024)

Skills

- **Surveying Tools:** Familiar in the use of Total Stations, Auto level, GNSS/GPS.
- **GIS Software:** Experience with ArcGIS, QGIS, ArcMap, Google Earth Engine for spatial data management, mapping and analysis.
- **Remote Sensing:** Skilled in processing satellite imagery for environmental monitoring and urban planning.
- **Data Analysis:** Competent in analyzing geospatial data to create detailed maps and reports.
- **Programming:** Working knowledge of Python for automating GIS processes and spatial data analysis.

- **Communication & Collaboration:** Excellent team player with strong written and verbal communication skills. Able to work effectively in interdisciplinary teams.
- **Fast learner:** Quickly grasps new concepts and technologies.
- **Willing to learn:** Proactive and highly motivated in gain industry experience in real world environment and continuously enhance technical and professional skills

EXPERIENCE

Academic Mentor

Tshwane University of Technology | January 2023 - November 2025

- Provided academic support and mentorship to junior Geomatics students, assisting with coursework, practical projects and exam preparation.
- Assisted students in **navigating university resources and building connections** with faculty and peers, creating a supportive learning environment.

Vacation Work

Thungela (Goedehoop Colliery Mine) | December 2024 - January 2025

- Conducted underground stability analysis to identify areas prone to subsidence and soil instability using risk based modelling and spatial analysis techniques.
- Utilized GIS-based mapping to identify and classify stable, moderately stable and unstable zones in the mine.
- Proposed rehabilitation strategies for unstable zones to ensure closure and future land use.

Student Assistant (Land surveying)

Tshwane University of Technology | February 2025 - November 2025

- Provide practical assistance to undergraduate students studying Geomatics and Civil engineering.
- Guide students in the use of surveying instruments during practical sessions.
- Promote a collaborative learning environment to improve student understanding and confidence in applied surveying techniques.

EXTRACURRICULAR ACTIVITIES

Geomatics Student Society executive member

Tshwane University of Technology | January 2023 - December 2024

- Participated in networking events with industry professionals, enhancing knowledge of the geomatics industry.
- Participated in **outreach initiatives** aimed at increasing awareness of geomatics as a field, inspiring interest in STEM among younger students.
- Collaborated with the team to **promote unity and engagement** among students, fostering a supportive environment for both academic and personal growth.

Robotic Skills Development Program (ROS)

Tshwane University of Technology | 27 November 2023 - 1 December 2023

- Gained hands-on experience in programming robots using ROS, a flexible framework for building robotic applications.
- Applied theoretical knowledge to create functional robotic systems for real-world applications.

REFERENCES

- Supervisor
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